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(54) **ANTENNA STRUCTURE**

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H01Q 21/26
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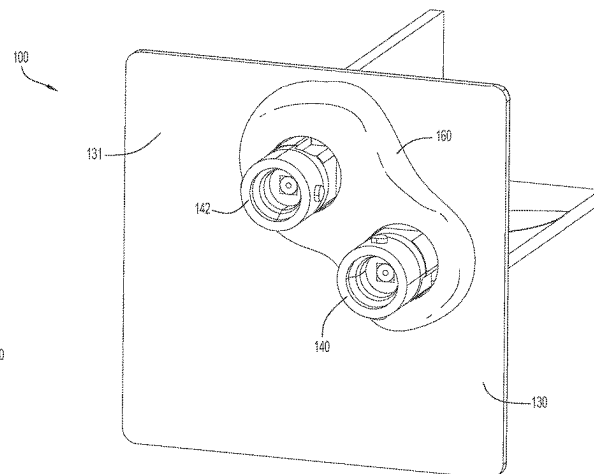
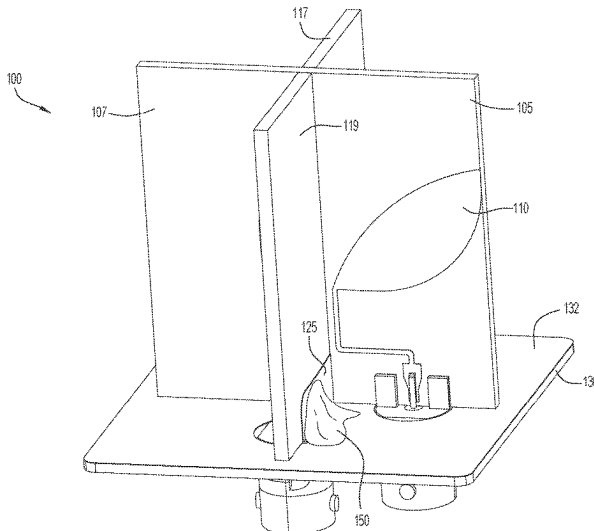
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(57) **ABSTRACT**

In some aspects, the techniques described herein relate to an
apparatus including: a ground plane element including: a
non-conductive support layer, a conductive layer arranged
on the non-conductive support layer, and at least one orifice
through the non-conductive support layer and the conductive
laminate layer; one or more radiating elements including a
feed line and a solder pad, wherein each of the one or more
radiating elements is secured to and electrically connected to
the ground plane element via soldering of the solder pad to
the conductive laminate layer; and at least one connector
arranged in the at least one orifice and electrically connected
to the feed line.

20 Claims, 15 Drawing Sheets



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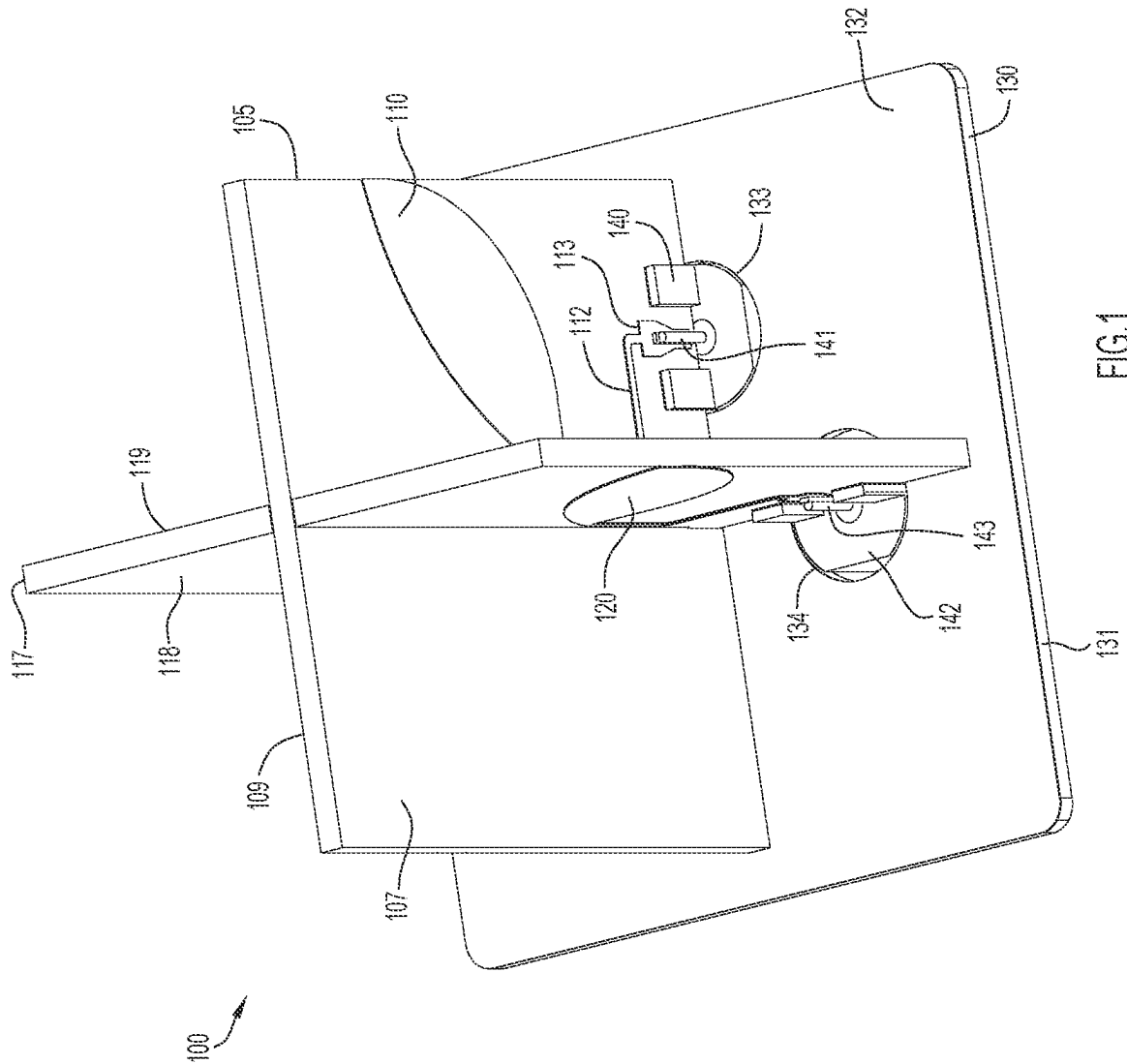
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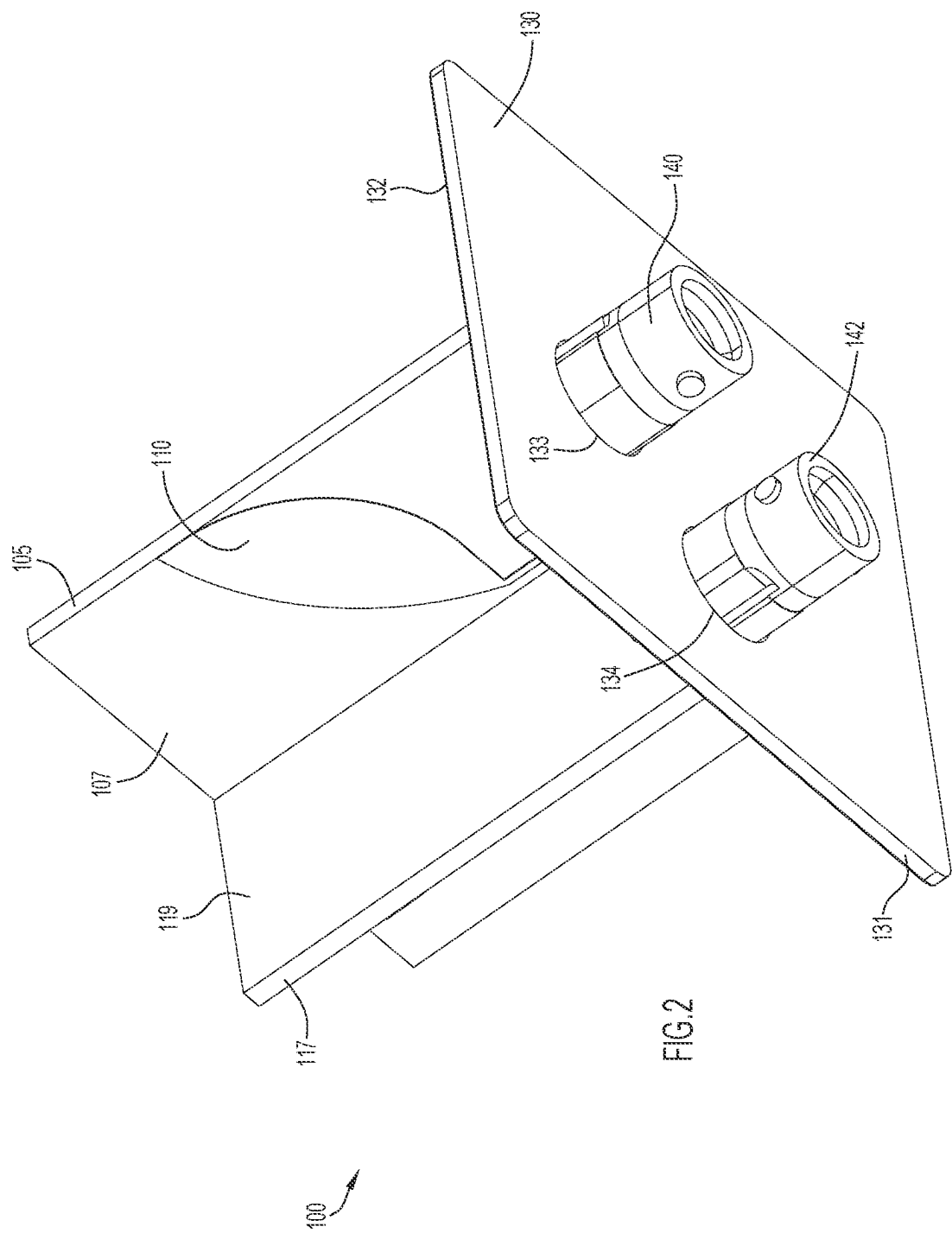
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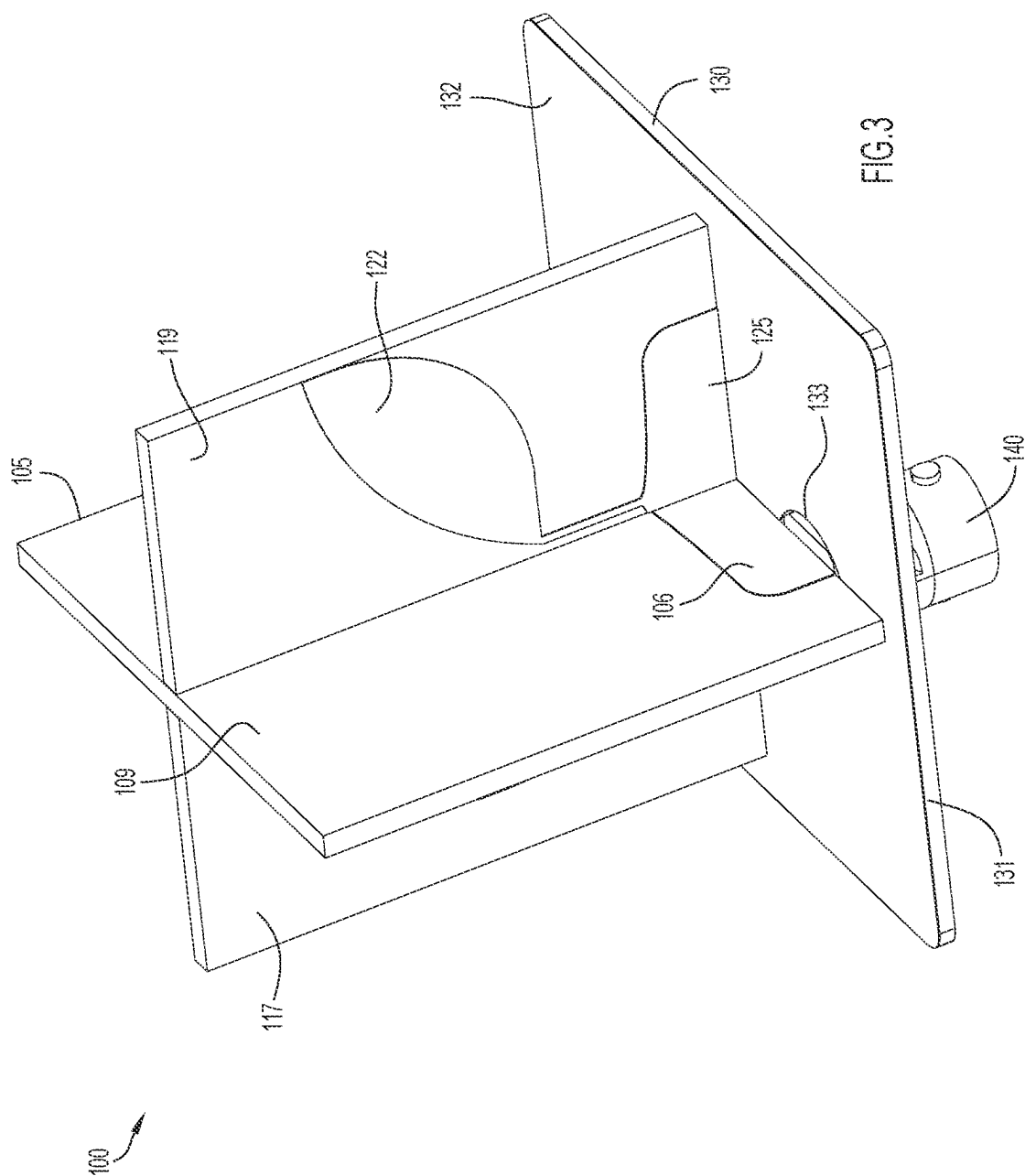
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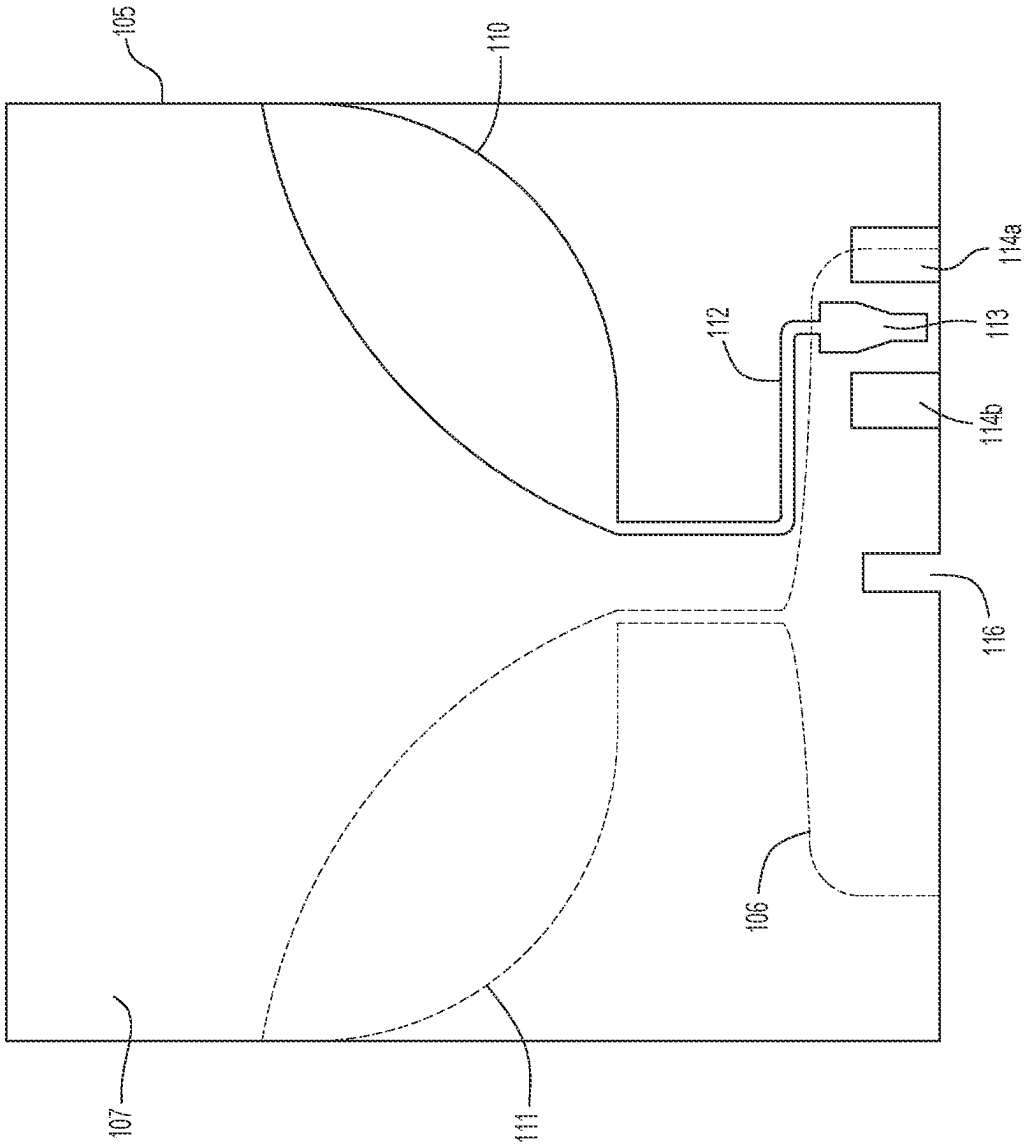


FIG. 4

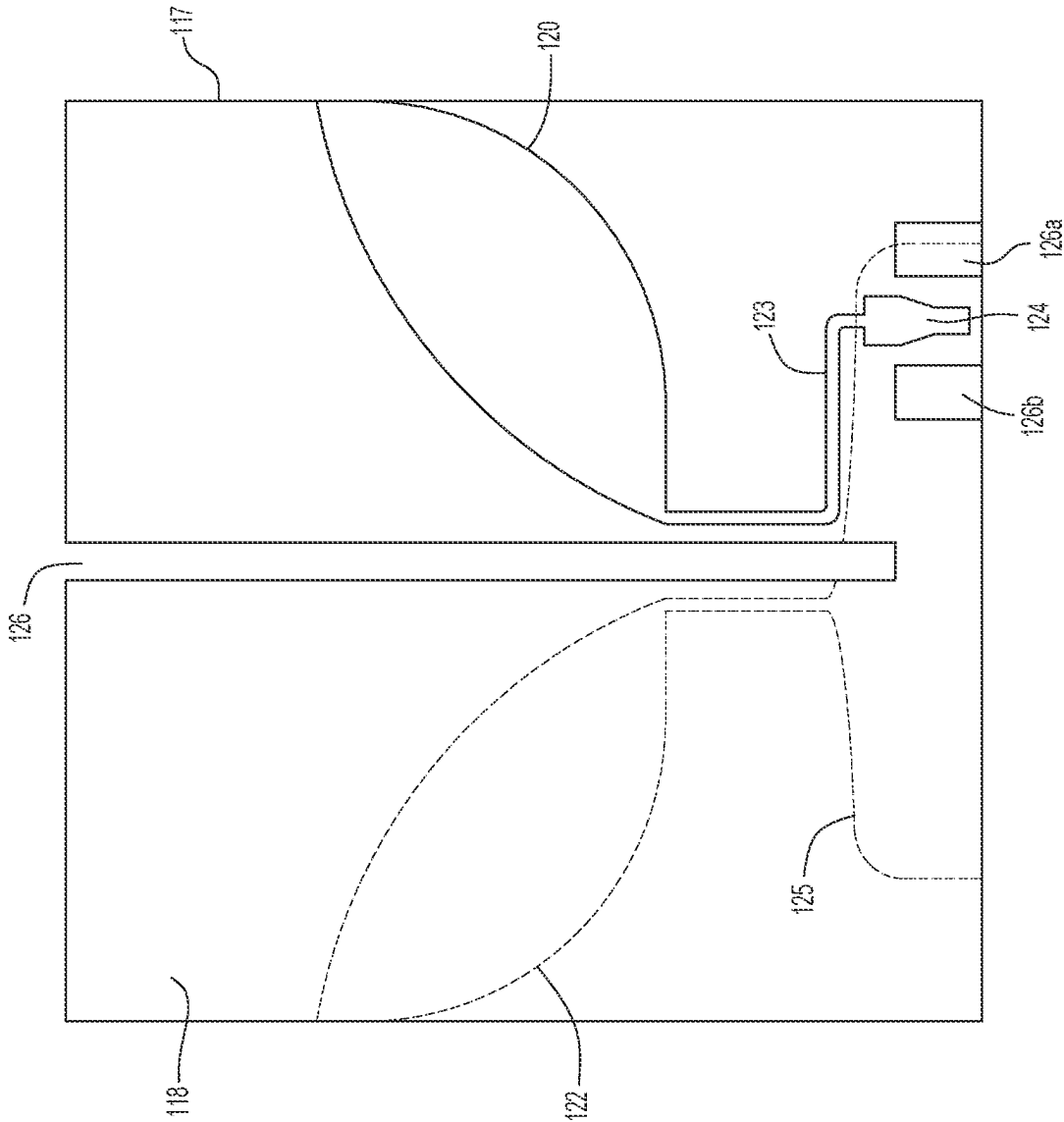


FIG. 5

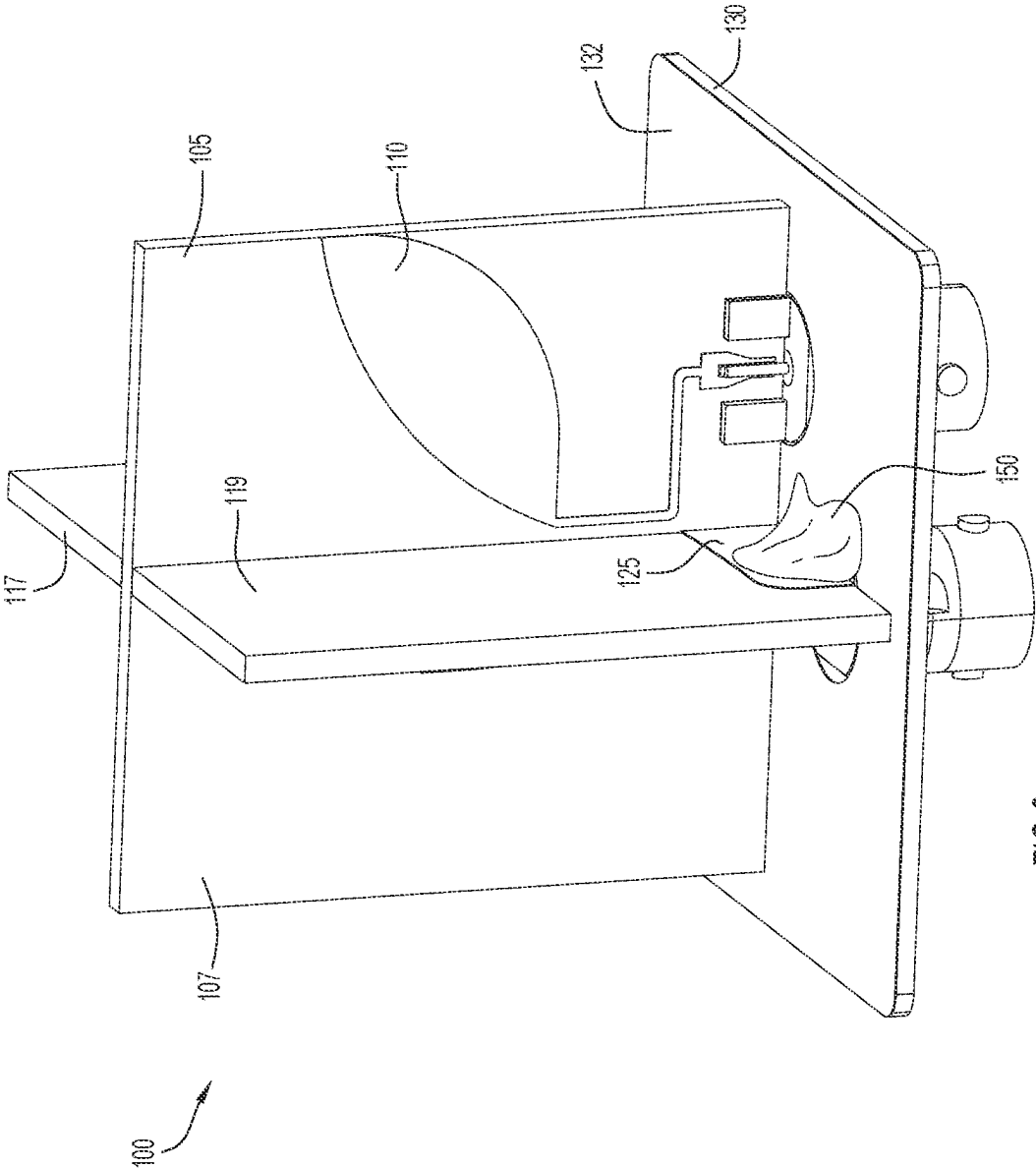


FIG. 6

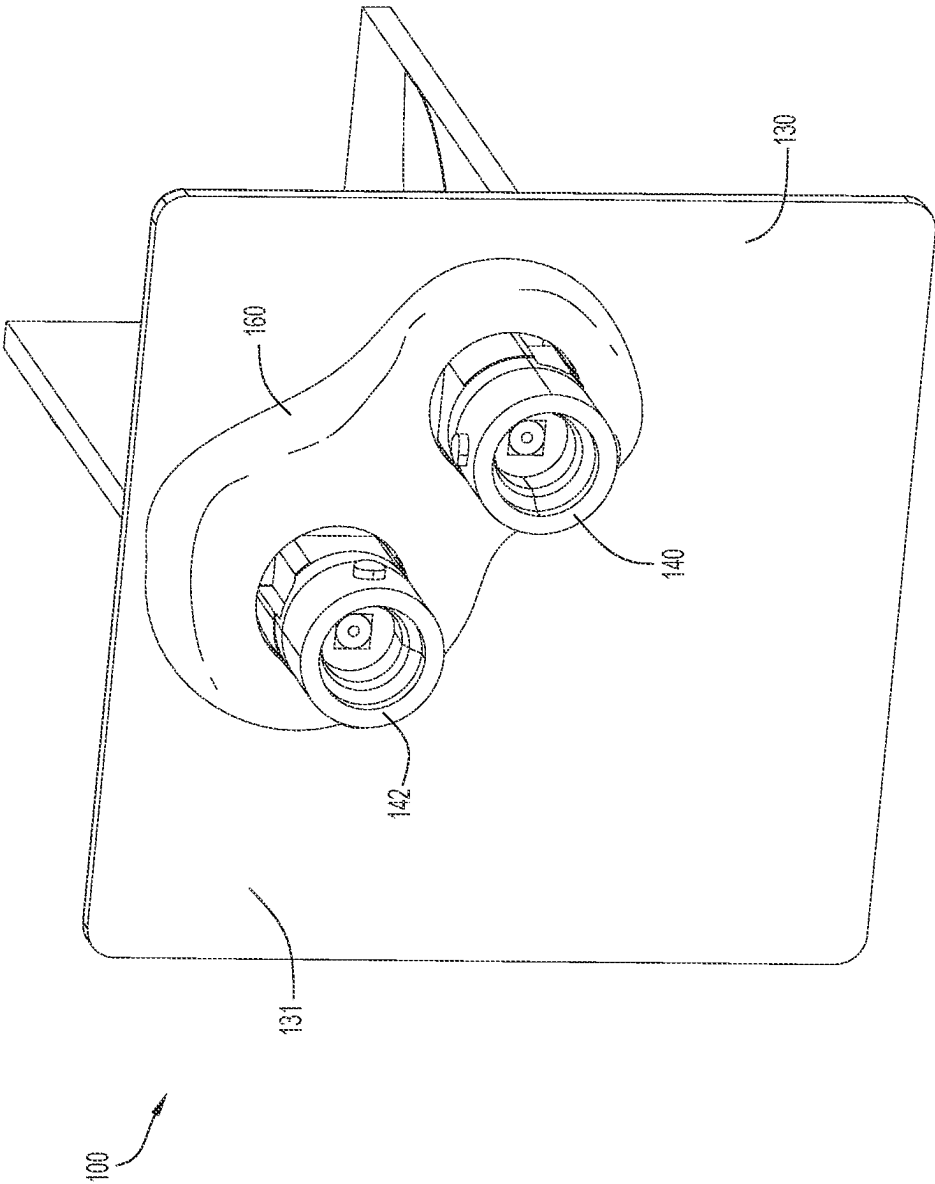
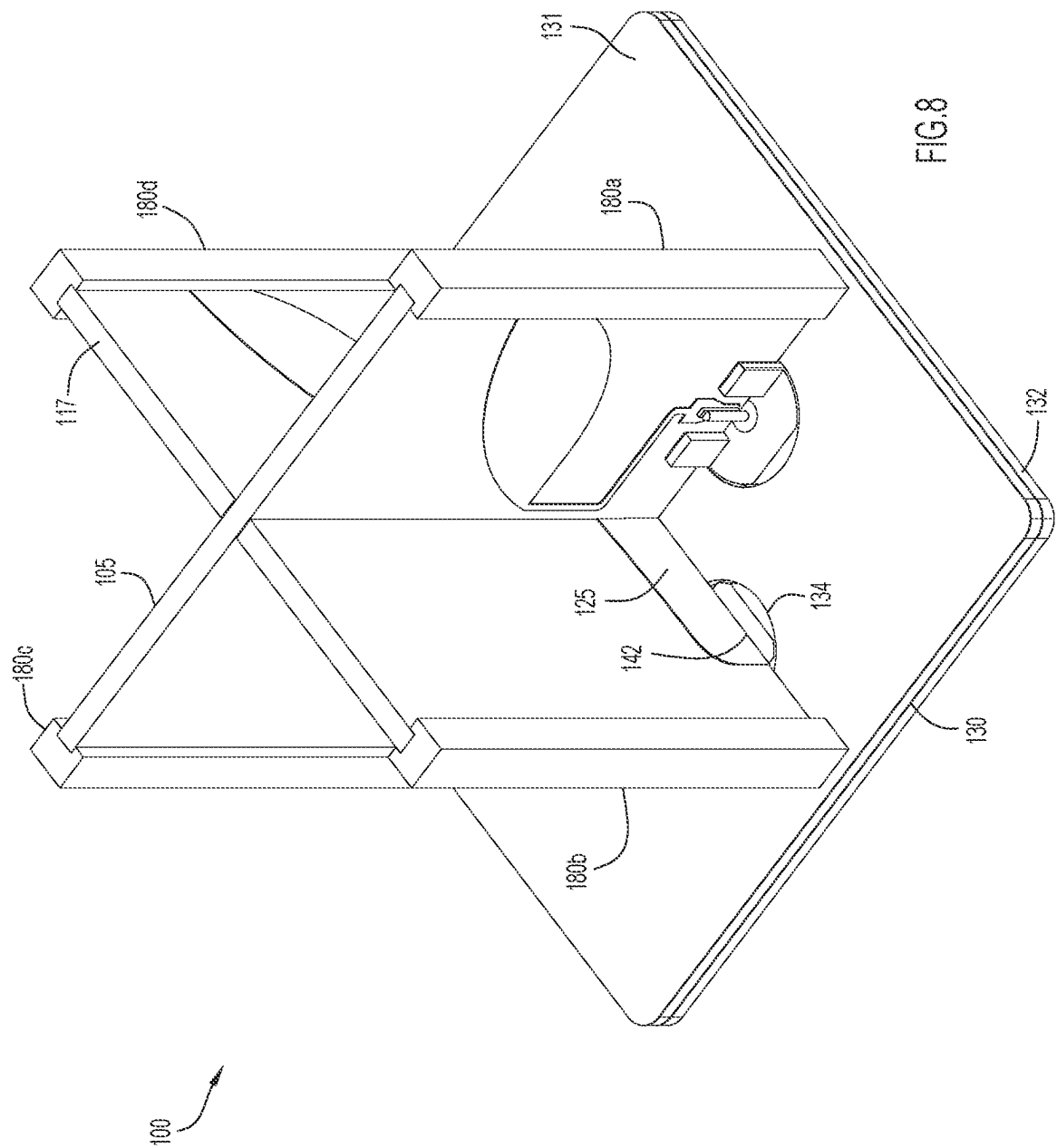
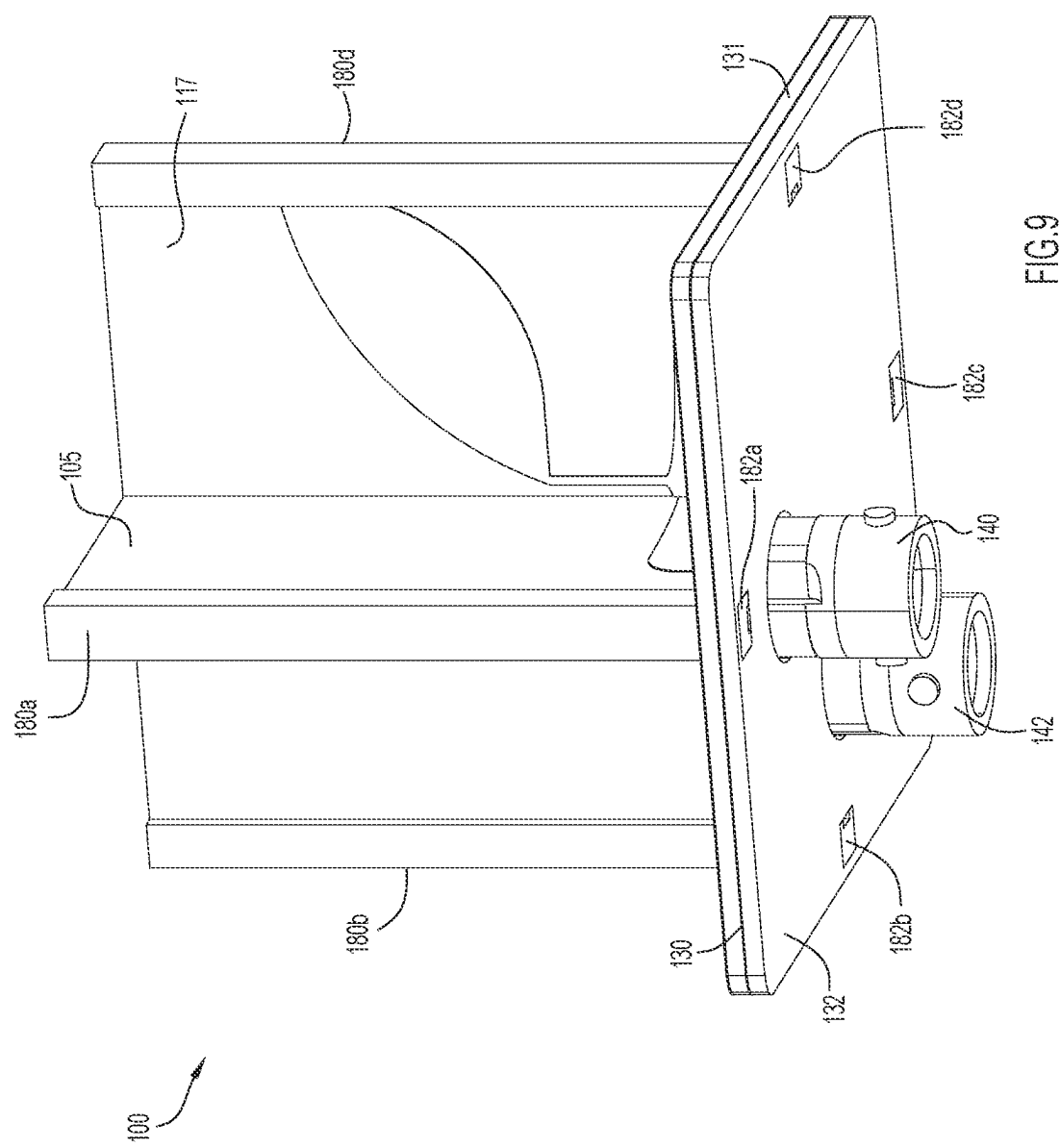


FIG. 7





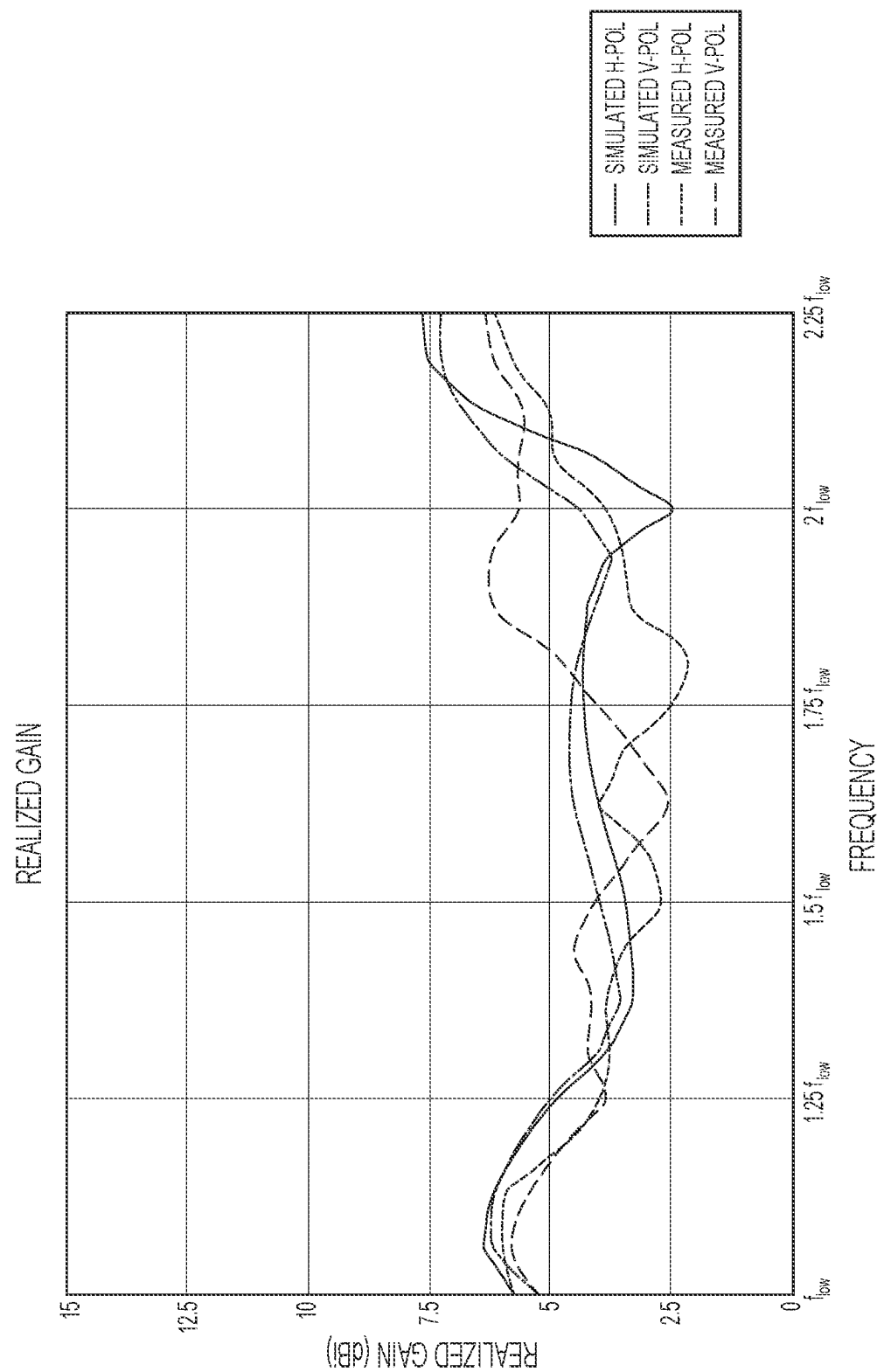


FIG.10

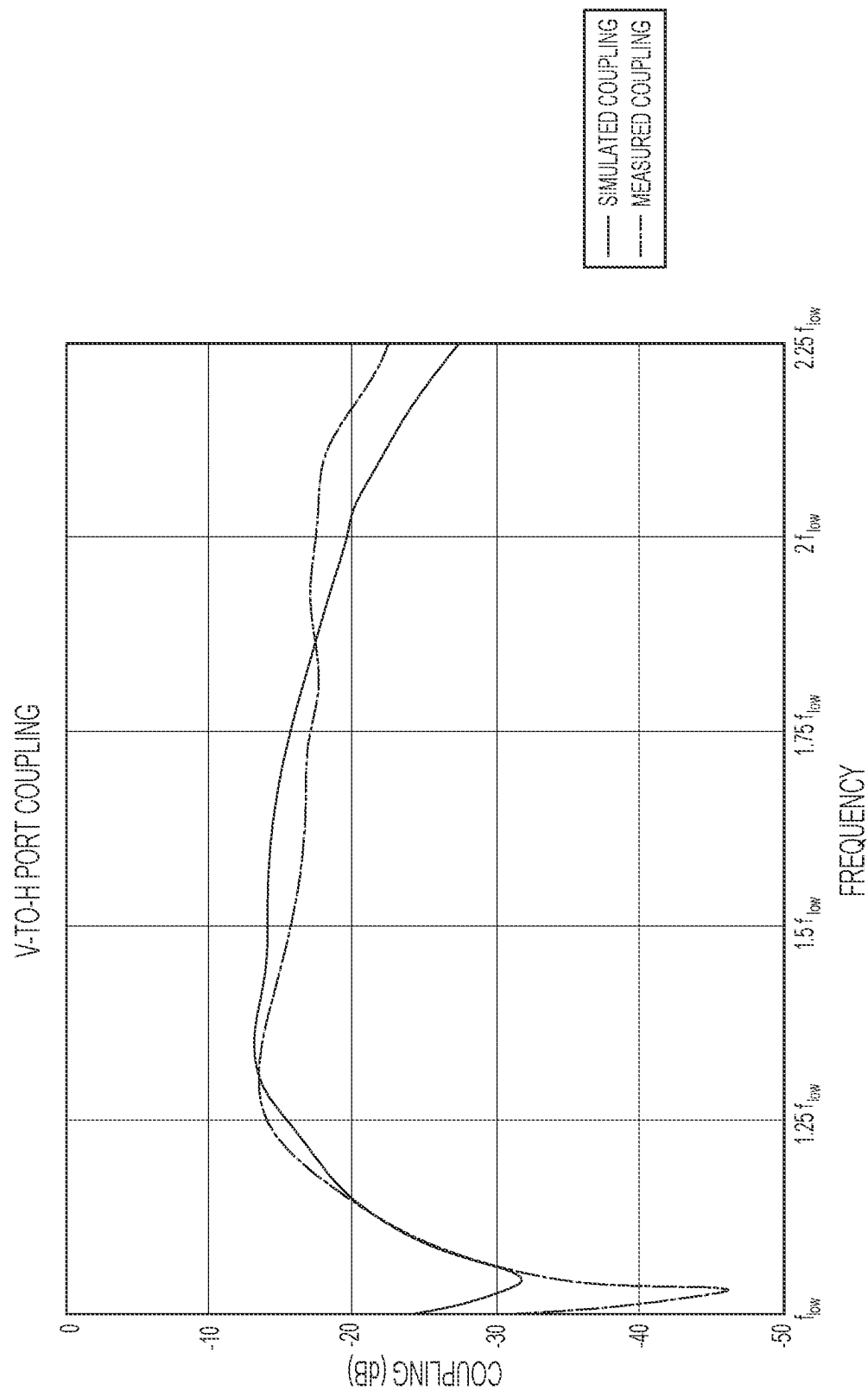


FIG.11

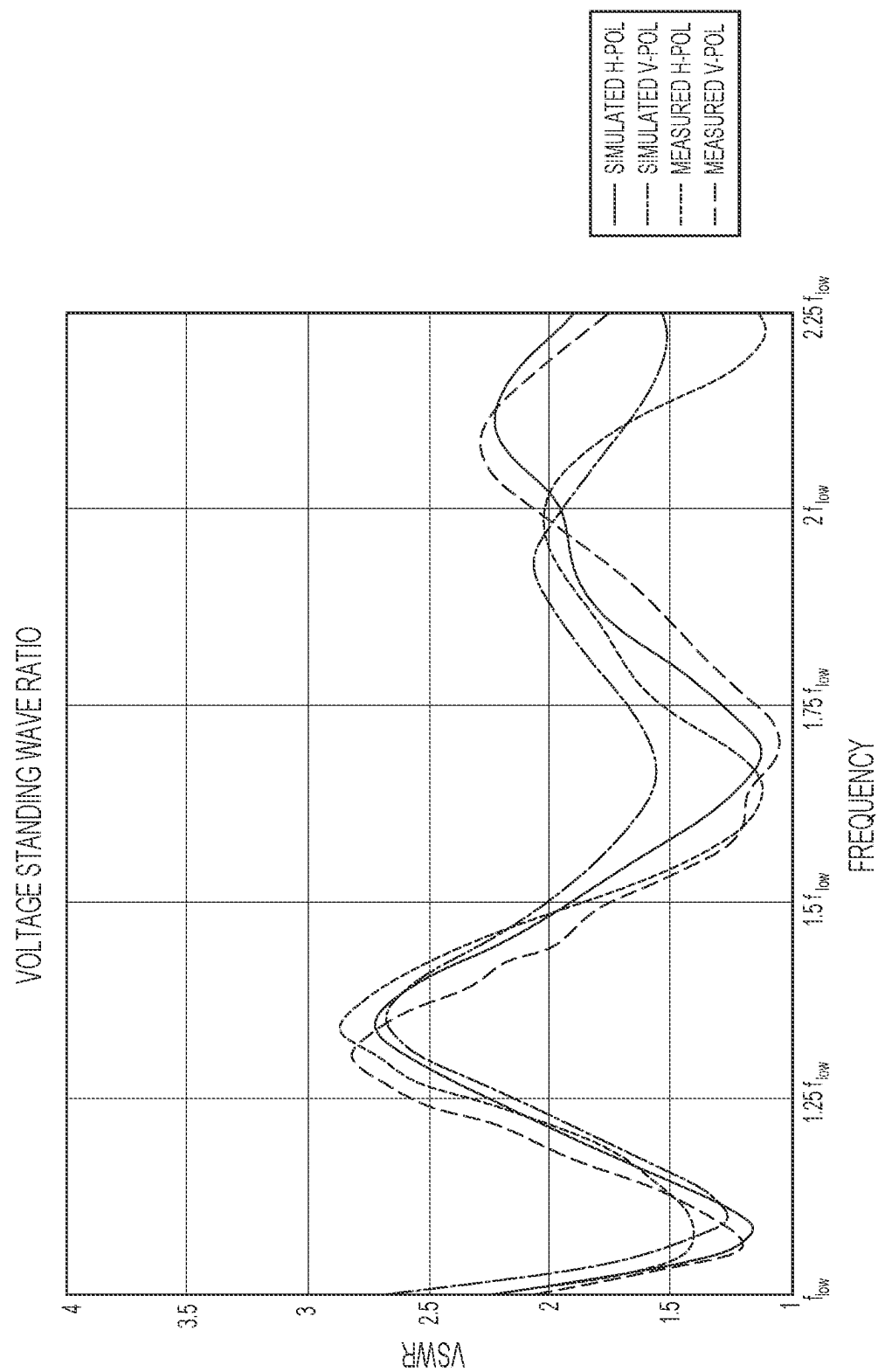
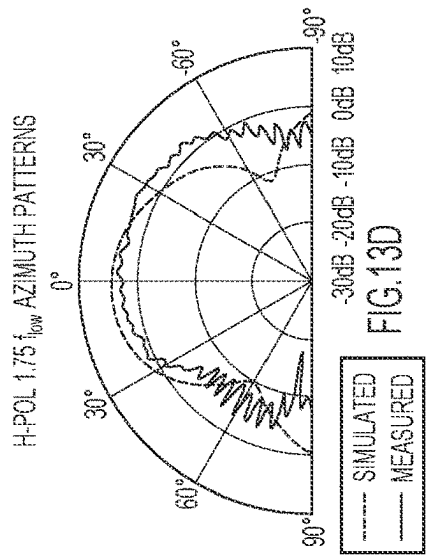
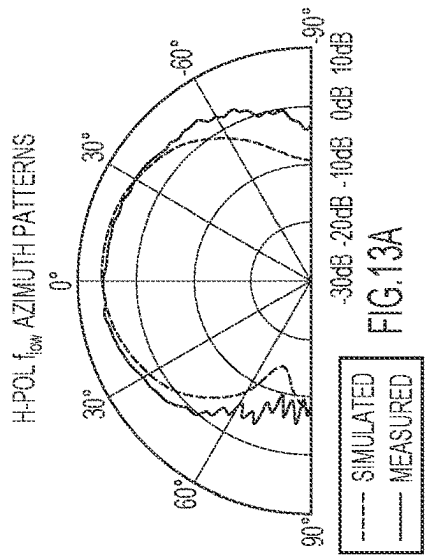
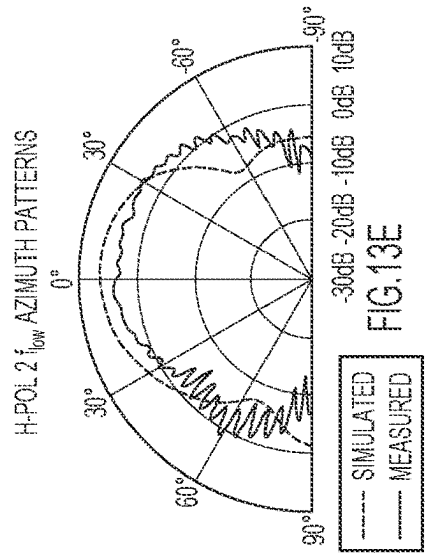
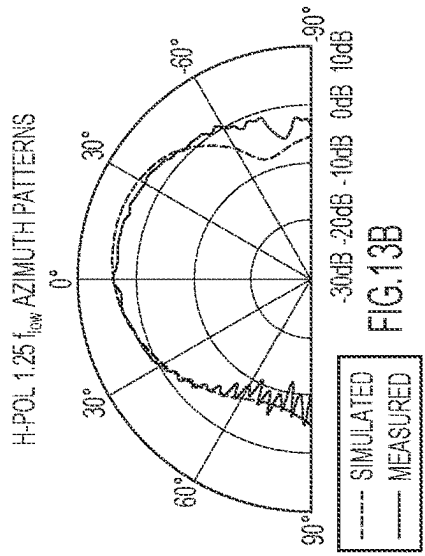
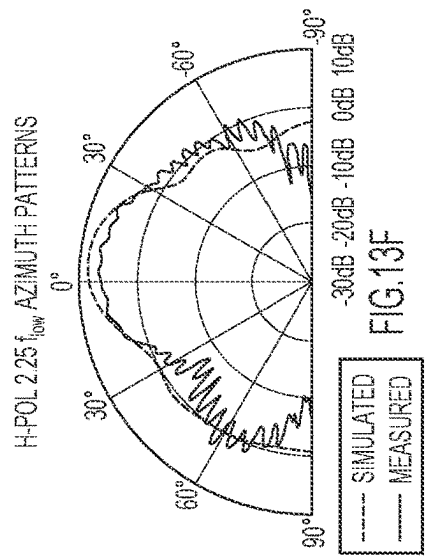
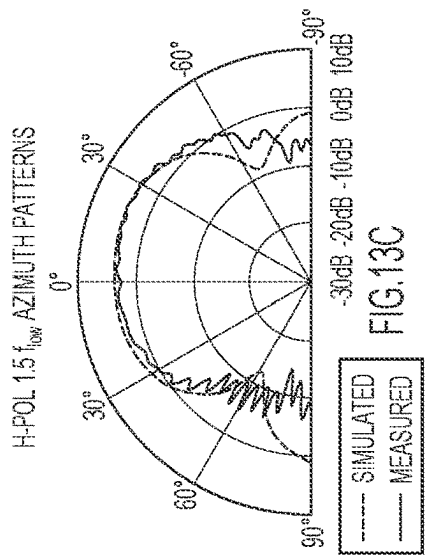
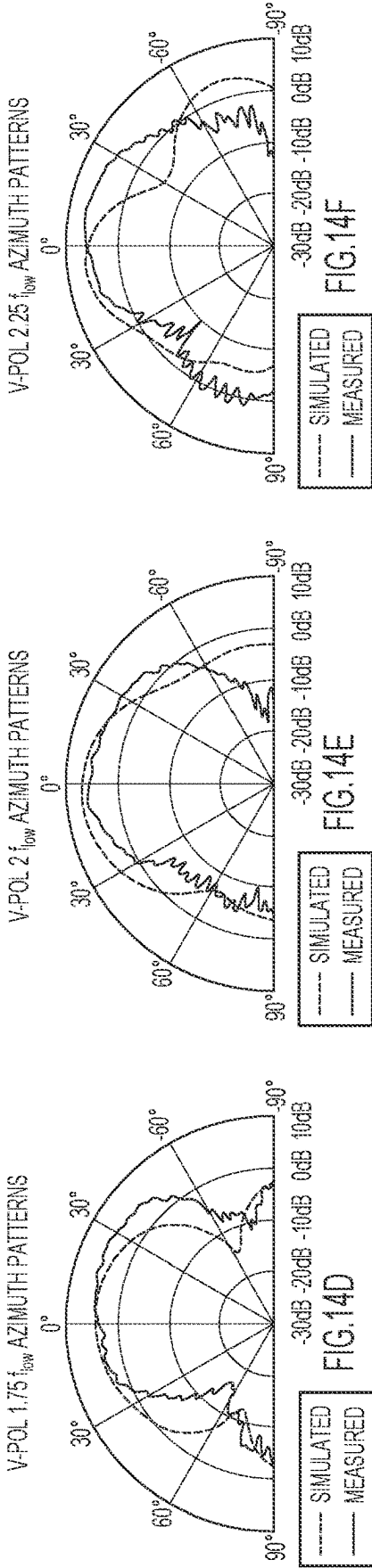
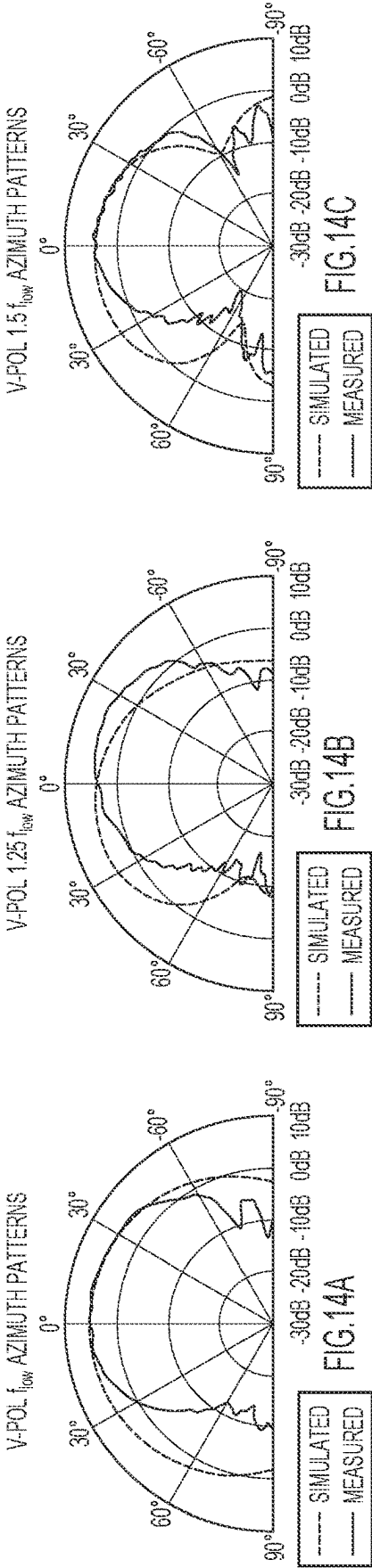
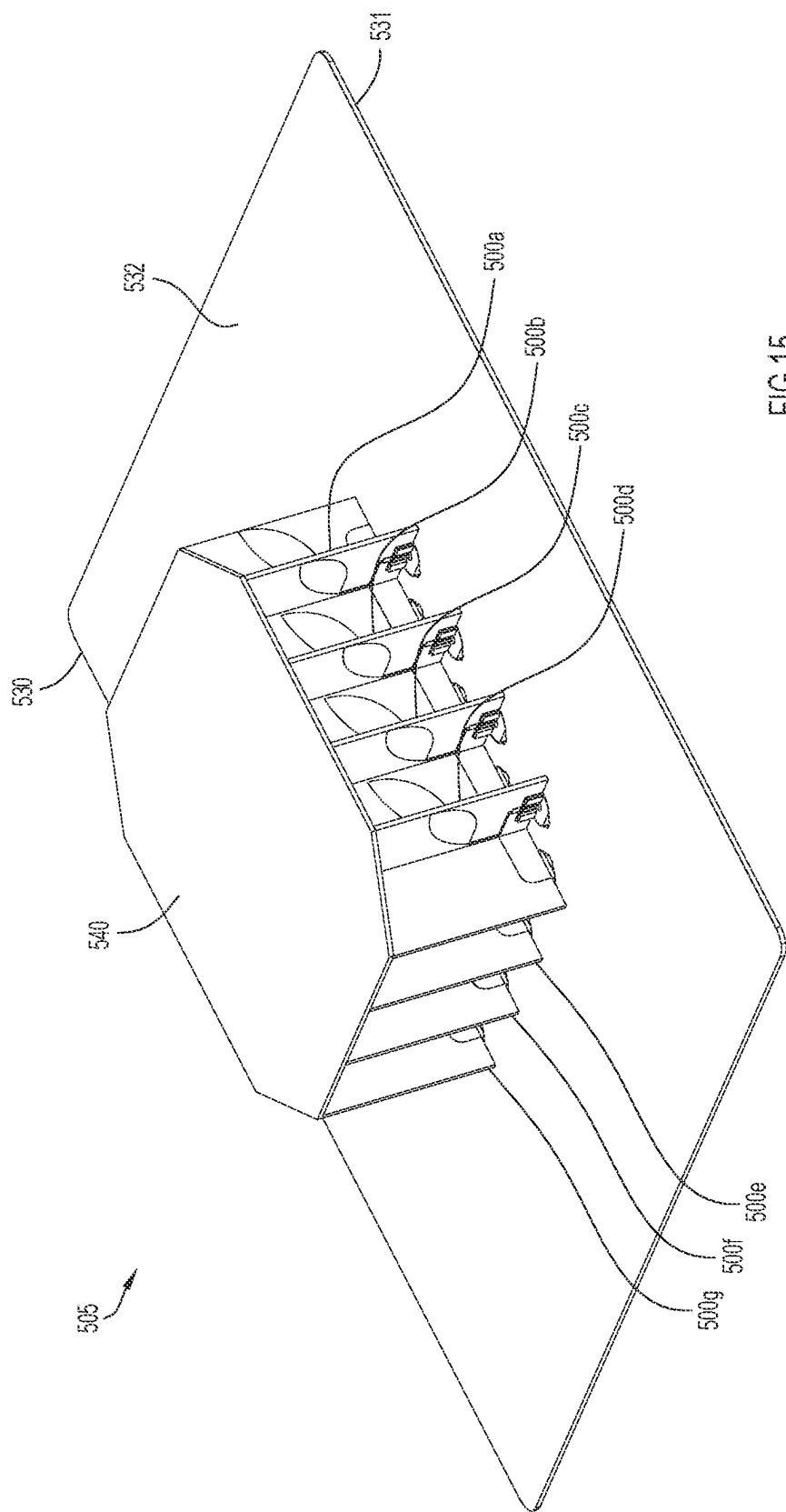


FIG.12







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ANTENNA STRUCTURE

TECHNICAL FIELD

The present disclosure relates to techniques for constructing antennas, including Vivaldi antennas.

BACKGROUND

A Vivaldi antenna (sometimes referred to as a tapered slot antenna) is traditionally a co-planar broadband-antenna. Vivaldi antennas may be constructed by forming an open space in a solid piece of sheet metal, a printed circuit board, or from a dielectric plate metalized on one or both sides. From the open space area the energy reaches an exponentially tapered pattern via a symmetrical slot line. In an antipodal Vivaldi antenna, a first Vivaldi arm is formed on a first side of a dielectric plate material, and a second Vivaldi arm is formed on a second side of the dielectric plate material. The open space is formed in areas of the dielectric plate material not covered by either radiating element. Vivaldi antennas may be made for linear polarized waves through the use of a single open space radiator. Using two radiators perpendicularly arranged allows for Vivaldi antennas that receive both horizontally and vertically polarized signals. If the arrangement is also concentric, the two radiating elements may be fed with 90-degree phase-shifted signals to provide circular polarized electromagnetic waves.

Vivaldi antennas may be configured for almost any frequency range, as the Vivaldi structure is scalable in size. Printed circuit technology makes this type of antenna cost effective at microwave frequencies exceeding 1 GHz. Advantages of Vivaldi antennas may include their broadband characteristics, their ease of manufacturing, and their ease of impedance matching using microstrip line modeling methods. However, dual-polarized Vivaldi antennas generally include heavy mechanical support structures.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a first perspective view of an antenna element constructed according to the techniques disclosed herein, according to an example embodiment.

FIG. 2 is a second perspective view of an antenna element constructed according to the disclosed techniques, according to an example embodiment.

FIG. 3 is a third perspective view of an antenna element constructed according to the disclosed techniques, according to an example embodiment.

FIG. 4 is a horizontal element of an antenna element constructed according to the disclosed techniques, according to an example embodiment.

FIG. 5 is a vertical element of an antenna element constructed according to the disclosed techniques, according to an example embodiment.

FIG. 6 illustrates the use of solder in constructing an antenna element according to the disclosed techniques, according to an example embodiment.

FIG. 7 illustrates the use of epoxy in constructing an antenna element according to the disclosed techniques, according to an example embodiment.

FIG. 8 is a first perspective view of an antenna element that includes support structures according to the disclosed techniques, according to an example embodiment.

FIG. 9 is a second perspective view of an antenna element that includes support structures according to the disclosed techniques, according to an example embodiment.

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FIG. 10 is a graph comparing measured and simulated realized gain for an antenna element constructed according to the disclosed techniques, according to an example embodiment.

FIG. 11 is a graph comparing measured and simulated vertical to horizontal port coupling for an antenna element constructed according to the disclosed techniques, according to an example embodiment.

FIG. 12 is a graph comparing measured and simulated voltage standing wave ratio (VSWR) for an antenna element constructed according to the disclosed techniques, according to an example embodiment.

FIGS. 13a-f contain graphs comparing measured and simulated horizontal radiation patterns for an antenna element constructed according to the disclosed techniques, according to an example embodiment.

FIGS. 14a-f contain graphs comparing measured and simulated vertical radiation patterns for an antenna element constructed according to the disclosed techniques, according to an example embodiment.

FIG. 15 is a perspective view of an antenna array constructed according to the techniques disclosed herein, according to an example embodiment.

DETAILED DESCRIPTION

Overview

In some aspects, the techniques described herein relate to an apparatus including: a ground plane element including: a non-conductive support layer, a conductive layer arranged on the non-conductive support layer, and at least one orifice through the non-conductive support layer and the conductive layer; one or more radiating elements including a feed line and a solder pad, wherein each of the one or more radiating elements is secured to and electrically connected to the ground plane element via soldering of the solder pad to the conductive layer; and at least one connector arranged in the at least one orifice and electrically connected to the feed line.

In some aspects, the techniques described herein relate to an apparatus including: a ground plane element including: a non-conductive support layer, a conductive layer arranged on the non-conductive support layer, and a plurality of orifices through the non-conductive support layer, and the conductive layer; a plurality of antenna elements arranged on the ground plane element, wherein each of the plurality of antenna elements includes a feed line and a solder pad, and wherein each of the plurality of antenna elements is secured to and electrically connected to the ground plane element via soldering of its respective solder pad to the conductive layer; and a plurality of connectors, wherein each of the plurality of connectors is arranged in a respective one of the plurality of orifices and electrically connected to a respective one of the feed lines.

Example Embodiments

In accordance with discussion above, provided for herein are techniques for providing wideband, lightweight, and low cost antenna elements. The antenna elements constructed according to the disclosed techniques may provide wideband radio frequency (RF) performance and polarization diverse capabilities from a single, light-weight antenna.

With reference made to FIGS. 1-5, depicted therein is an antipodal Vivaldi antenna (AVA) 100 configured according to the techniques disclosed herein. AVA 100 includes a horizontal element 105, a vertical element 117 and a ground

plane element **130**. As an antipodal Vivaldi antenna, horizontal element **105** includes a first side **107** and a second side **109**, illustrated in more detail in FIG. 4. As shown in FIG. 4, a first Vivaldi arm **110** is arranged on first side **107** and a second Vivaldi arm **111** is arranged on the second side **109**. The first Vivaldi arm **110** and the second Vivaldi arm **111** are formed as microstrip structures on the copper clad laminate substrate of horizontal element **105**. The microstrip structure of first Vivaldi arm **110** includes feed line connection pad **113** which connects feed line **112** to antenna feed pin **141** (illustrated in FIG. 1). Second Vivaldi arm **111** is formed as a microstrip structure on second side **109** of horizontal element **105**. Included in second Vivaldi arm **111** is solder pad **106**. As will be described in greater detail below, solder pad **106** will be used to solder horizontal element **105** to ground plane element **130**. Horizontal element **105** also includes solder pads **114a** and **114b** for attaching sub miniature push-on micro (SMPM) connector **140**. Horizontal element **105** also includes notch **116** which engages with channel **126** of vertical element **117** (illustrated in FIG. 5).

Turning to vertical element **117** (illustrated in detail in FIG. 5), vertical element **117** includes a similar microstrip structure as that of horizontal element **105**. Vertical element **117** includes a first side **118** and a second side **119**. As shown in FIG. 5, a first vertical Vivaldi arm **120** is arranged on first side **118** and a second vertical Vivaldi arm **122** is arranged on the second side **119**. The first vertical Vivaldi arm **120** and the second vertical Vivaldi arm **122** are formed as microstrip structures on the copper clad laminate substrate of vertical element **117**. The microstrip structure of first vertical Vivaldi arm **120** includes feed line connection pad **124** which connects feed line **123** to antenna feed pin **143** (illustrated in FIG. 1). Second vertical Vivaldi arm **122** is formed as a microstrip structure on second side **119** of vertical element **117**. Included in second vertical Vivaldi arm **122** is solder pad **125**. As will be described in greater detail below, solder pad **125** will be used to solder vertical element **117** to ground plane element **130**. Vertical element **117** includes solder pads **126a** and **126b** for attaching SMPM connector **142**. As noted above, vertical element **117** also includes channel **126** which engages with notch **116** of horizontal element **105** (illustrated in FIG. 4).

Horizontal element **105** and vertical element **117** are arranged on ground plane element **130**, as illustrated in FIGS. 1-3. According to the techniques disclosed herein, ground plane element **130** is formed from a combination of a non-conductive support layer **131** and a conductive layer **132**. According to the specific example embodiment of FIGS. 1-3, ground plane element **130** is formed from a laminate of copper (serving as the conductive layer **132**) and a dielectric or insulating material (serving as the non-conductive support layer **131**) which are laminated together to form a copper laminate material. According to a specific example, the copper laminate material may be constructed from a printed circuit board (PCB) material, in which the copper foil serves as the conductive layer **132** and the PCB core serves as the non-conductive support layer. Ground plane element **130** may include additional materials, such as a PCB prepreg layer.

According to another example, the conductive layer **132** may be constructed from a spun metal layer, and the non-conductive support layer **131** may be constructed from a polyimide material. As understood by the skilled artisan, polyimide materials are polymers containing imide groups

belonging to the class of high-performance plastics, such as those used in power electronics found in high speed switch devices.

As understood from the example materials described above, conductive layer **132** may be formed from a thin conductive layer, such as a copper or other metallic foil or a thin spun metal layer. The use of such materials for ground plane element **130** enables AVA **100** to have a lightweight design while retaining desirable performance characteristics. Specifically, ground plane element **130** may provide a lightweight ground plane with strong radio frequency (RF) performance. Typical related art Vivaldi antennas rely on heavy aluminum mechanical structures to achieve similar RF performance. Accordingly, AVA **100** maintains similar RF performance with significantly lighter weight mechanical structures.

Ground plane element **130** includes two orifices-orifice **133** and orifice **134**. Orifice **133** allows SMPM connector **140** to pass through ground plane element **130** and connects antenna feed pin **141** to feed line connection pad **113**. Orifice **134** allows SMPM connector **142** to pass through ground plane element **130** and connects antenna feed pin **143** to feed line connection pad **124**.

To secure horizontal element **105** to ground plane element **130**, solder pad **106** is soldered to conductive layer **132**. Similarly, vertical element **117** is secured to substrate **130** by soldering solder pad **125** to conductive layer **132**. The use of solder **150** to connect vertical element **117** to conductive layer **132** is illustrated in FIG. 6. While obscured in FIG. 6 by horizontal element **105**, horizontal element **105** is similarly secured to ground plane element **130** via solder between solder pad **106** and conductive layer **132**.

Further structural integrity is provided to AVA **100** by epoxying SMPM connectors **140** and **142** to non-conductive support layer **131** of ground plane element **130**, as illustrated in FIG. 7. As illustrated in FIG. 7, epoxy **160** secures SMPM connectors **140** and **142** to non-conductive support layer **131**. The combination of solder **150** (illustrated in FIG. 6) and epoxy **160** (illustrated in FIG. 7) provide for a structurally sound antenna structure that is simple to construct, easy to integrate into, for example, antenna arrays, and that is easily scalable to operate at the L (1-2 GHz), S (2-4 GHz), C (4-8 GHz), X (8-12 GHz), Ku (12-18 GHz), K (18-27 GHz) and Ka (27-40 GHz) frequency bands.

Based upon the feeding provided by SMPM connector **140** and SMPM connector **142**, AVA **100** may radiate linear horizontal polarization by feeding only SMPM connector **140**, radiate linear vertical polarization by feeding only SMPM connector **142**, and radiate dual-circular polarization by connecting to a 90° hybrid coupler to provide a 90° phase shift between SMPM connector **140** and SMPM connector **142**.

As discussed above, the combination of solder **150** (illustrated in FIG. 6) and epoxy **160** (illustrated in FIG. 7) provide for a structurally sound antenna structure. However, if additional structural integrity is required or beneficial to a particular application, additional support structures may be included in AVA **100**, as illustrated in FIGS. 8 and 9. As shown in FIGS. 8 and 9, support structures **180a-d** may be included in AVA **100**. Support structures **180a** and **180c** may be provided with a "U" shape cross-section to engage with the ends of horizontal element **105**. Similarly, support structures **180b** and **180d** are configured to engage with the ends of vertical element **117**. As illustrated in FIG. 9, ground plane element **130** includes orifices **182a-d** formed in non-conductive support layer **131** and conductive layer **132** to engage and secure support structures **180a-d**. Orifices

180a-d may be sized such that they secure support structures **180a-d** via a friction fit. According to other example embodiments support structures **180a-d** may be secured in orifices **182a-d** via gluing, epoxying, sonic welding, or techniques known to the skilled artisan.

FIGS. **8** and **9** also differ from the embodiment of FIGS. **1-5** in that the orientation of non-conductive support layer **131** and conductive layer **132** is reversed. Specifically, non-conductive support layer **131** is on the same side of the ground plane element **130** as horizontal element **105** and the vertical element **117**. Accordingly, the embodiment of FIGS. **8** and **9** illustrates an alternative method of mounting horizontal element **105** and the vertical element **117** to ground plane element **130**. According to this embodiment, solder pad **125** would be soldered to the portion of connector **142** exposed by orifice **134**. Connector **142** would then be soldered to conductive layer **132** on the opposite side of ground plane element **130** electrically and mechanically connecting vertical element **117** to ground plane element **130**. Connector **140** would be analogously soldered to horizontal element **105** and conductive layer **132** to electrically and mechanically connect horizontal element **105** to ground plane element **130**. This reversed ground plane element **130** may be applied to the structure illustrated in FIGS. **1-7** (i.e., the embodiment without support structures **180a-d** and orifices **182a-d**) without deviating the antenna construction techniques disclosed herein.

In addition to the structural benefits discussed above, antennas constructed according to the disclosed techniques also exhibit beneficial electrical properties. For example, illustrated in FIG. **10** is a comparison of the realized gain of an AVA constructed according to the disclosed techniques compared to the simulated values. Specifically, FIG. **10** compares the simulated and measured realized gain for both horizontal and vertical polarized operation of the antenna element. FIG. **11** provides a graph comparing the measured versus simulated vertical-to-horizontal port coupling of the AVA. FIG. **12** provides a graph comparing the measured versus simulated voltage standing wave ratio for both horizontal and vertical polarized operation of the AVA. As illustrated in these graphs, the AVA element not only exhibits strong electrical performance, but its measured performance substantially matches its simulated performance. This agreement between the measure and simulated performance evidences the success of the disclosed techniques. Because the measured performance substantially matches the simulated performance, the techniques may be confidently applied when designing new antenna structures knowing the resulting physical antenna will behave as expected.

The disclosed techniques also provide for radiation patterns that are not only applicable to many different applications, but that also match their simulated performance. Illustrated in FIGS. **13a-f** are comparisons of the measured horizontal radiation pattern for an AVA constructed according to the disclosed techniques compared with the simulated pattern for the same structure, across an operating frequency range of the antenna. Not only are the radiation patterns fairly hemispherical, but the measured patterns substantially match the simulation patterns. FIGS. **14a-f** illustrate comparisons of the measured vertical radiation pattern for an AVA constructed according to the disclosed techniques compared with the simulated pattern for the same structure across the operating frequency range of the antenna. As with the horizontal radiation patterns of FIGS. **13a-f**, the vertical radiation patterns of FIGS. **14a-f** are fairly hemispherical and substantially match the simulation patterns.

In summary, provided herein are techniques for providing an antenna and antenna structure that has a simple and light-weight construction. The disclosed techniques also provide for antennas that exhibit desirable gain, radiation patterns, and S-parameters in a small package. Furthermore, testing of antennas constructed according to the disclosed techniques exhibit performance that closely measures their simulated performance, indicating a successful and repeatable antenna design. Given the lightweight and high gain performance achievable through the disclosed techniques, the techniques may be particularly applicable to airborne applications, including satellite low earth orbit applications. When the disclosed techniques are leveraged to implement AVA elements, the techniques may be particularly applicable to airborne telemetry applications.

The techniques of this disclosure are not limited to antenna elements. Instead, as illustrated in FIG. **15**, the disclosed techniques may be applied to antenna arrays. Specifically, illustrated in FIG. **15** are a number of AVA antenna elements **500a-g** that have been arranged into antenna array **505**. While only seven AVA antenna elements **500a-g** are visible, antenna array **505** is 4x4 array that includes 16 AVA antenna elements. Each of antenna elements **500a-g** is constructed substantially the same as AVA **100** of FIGS. **1-7**. However, antenna array **505** includes a single ground plane element **530**. Like ground plane element **130**, ground plane element **530** is constructed from non-conductive support layer **531** and a conductive layer **532**. Accordingly, ground plane element **530** may be formed from a laminate of copper (serving as the conductive layer **532**) and a non-conductive material (serving as the non-conductive support layer **531**) which are laminated together to form a copper laminate material. Non-conductive support layer **531** provides a light substrate that supports AVA antenna elements **500a-g**, while conductive layer **532** exhibits the necessary electrical properties so that ground plane element **530** serves as an effective ground plane for antenna array **505**. Antenna array **505** may also include a cover **540** which serves to protect and add structural rigidity to antenna array **505**. Cover **540** is configured to be essentially transparent to the radiation frequencies that are sent or received by antenna elements **500a-g** so as to not negatively affect the performance of antenna array **505**. For example, cover **540** may be constructed from a closed cell foam. A closed cell foam layer may also be added to ground plane element **130** provide additional structural support and/or rigidity to antenna array **505**.

The above description of AVA **100** and antenna array **505** has been provided with regard to an example embodiment in which the radiating elements are embodied as Vivaldi elements, and in particular AVA elements. However, the techniques described herein may be applied to other types of antenna elements, including vertical tightly coupled dipole antenna elements and arrays, Planar Ultrawideband Modular Antenna (PUMA) element and arrays, and Stacked Patch elements and arrays. As understood by the skilled artisan from the present disclosure, the disclosed techniques may be implementing in conjunction with any number of different antenna elements to significantly reduce the weight, design complexity, cost, and schedule for designing and constructing antenna elements and arrays.

In summary, the techniques described herein relate to an apparatus including: a ground plane element including: a non-conductive support layer, a conductive layer arranged on the non-conductive support layer, and at least one orifice through the non-conductive support layer and the conductive layer; one or more radiating elements including a feed line

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and a solder pad, wherein each of the one or more radiating elements is secured to and electrically connected to the ground plane element via soldering of the solder pad to the conductive layer; and at least one connector arranged in the at least one orifice and electrically connected to the feed line.

In some aspects, the techniques described herein relate to an apparatus, wherein the one or more radiating elements includes a Vivaldi radiating element.

In some aspects, the techniques described herein relate to an apparatus, wherein the Vivaldi radiating element includes an antipodal Vivaldi radiating element.

In some aspects, the techniques described herein relate to an apparatus, wherein the one or more radiating elements include a first Vivaldi radiating element and a second Vivaldi radiating element arranged orthogonally and concentrically to each other.

In some aspects, the techniques described herein relate to an apparatus, wherein the first Vivaldi radiating element is configured to transmit or receive radiation with a first linear polarization; and wherein the second Vivaldi radiating element is configured to transmit or receive radiation with a second linear polarization orthogonal to the first linear polarization.

In some aspects, the techniques described herein relate to an apparatus, wherein the first Vivaldi radiating element and the second Vivaldi radiating element are configured to transmit or receive radiation with circular polarization.

In some aspects, the techniques described herein relate to an apparatus, wherein the conductive layer includes a copper conductive laminate layer.

In some aspects, the techniques described herein relate to an apparatus, wherein the non-conductive support layer and the conductive layer include a copper clad laminate material.

In some aspects, the techniques described herein relate to an apparatus, wherein the at least one connector is epoxied to the ground plane element.

In some aspects, the techniques described herein relate to an apparatus, further including a support structure secured within a second orifice in the ground plane element and supporting an edge of the one or more radiating elements.

In some aspects, the techniques described herein relate to an apparatus, wherein the one or more radiating elements includes a microstrip structure on a copper clad laminate substrate.

In some aspects, the techniques described herein relate to an apparatus, wherein the solder pad includes a microstrip structure on a copper clad laminate substrate.

In some aspects, the techniques described herein relate to an apparatus including: a ground plane element including: a non-conductive support layer, a conductive layer arranged on the non-conductive support layer, and a plurality of orifices through the non-conductive support layer, and the conductive layer; a plurality of antenna elements arranged on the ground plane element, wherein each of the plurality of antenna elements includes a feed line and a solder pad, and wherein each of the plurality of antenna elements is secured to and electrically connected to the ground plane element via soldering of its respective solder pad to the conductive layer; and a plurality of connectors, wherein each of the plurality of connectors is arranged in a respective one of the plurality of orifices and electrically connected to a respective one of the feed lines.

In some aspects, the techniques described herein relate to an apparatus, wherein each of the plurality of antenna elements includes a Vivaldi antenna element.

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In some aspects, the techniques described herein relate to an apparatus, wherein each of the plurality of antenna elements includes an antipodal Vivaldi antenna element.

In some aspects, the techniques described herein relate to an apparatus, wherein the conductive layer includes a copper laminate layer.

In some aspects, the techniques described herein relate to an apparatus, wherein each of the plurality of connectors is epoxied to the non-conductive support layer.

In some aspects, the techniques described herein relate to an apparatus, further including a cover arranged on the plurality antenna elements opposite to the ground plane element.

In some aspects, the techniques described herein relate to an apparatus, wherein the cover is transparent to radiation sent or received by the plurality of antenna elements.

In some aspects, the techniques described herein relate to an apparatus, wherein the cover includes a closed cell foam.

The above description is intended by way of example only. Although the techniques are illustrated and described herein as embodied in one or more specific examples, it is nevertheless not intended to be limited to the details shown, since various modifications and structural changes may be made within the scope and range of equivalents of the claims.

What is claimed is:

1. An apparatus comprising:

a ground plane element comprising:

a non-conductive support layer,
a conductive layer arranged on the non-conductive support layer, and
a first orifice through the non-conductive support layer and the conductive layer, and
a second orifice through the non-conductive support layer and the conductive layer;

a first Vivaldi radiating element comprising a first feed line and a first solder pad, wherein the first Vivaldi radiating element is electrically connected to the ground plane element via first soldering of the first solder pad to the conductive layer;

a second Vivaldi radiating element comprising a second feed line and a second solder pad, wherein the second Vivaldi radiating element is electrically connected to the ground plane element via second soldering of the second solder pad to the conductive layer, wherein the second Vivaldi radiating element is arranged orthogonally and concentrically relative to the first Vivaldi radiating element;

a first connector arranged in the first orifice and electrically connected to the first feed line;

a second connector arranged in the second orifice and electrically connected to the second feed line;

first epoxy epoxying the first connector to the non-conductive support layer; and

second epoxy epoxying the second connector to the non-conductive support layer,

wherein the first soldering and first epoxy secure the first Vivaldi radiating element to the non-conductive support layer and the second soldering and second epoxy secure the second Vivaldi radiating element to the non-conductive support layer without further structural supports.

2. The apparatus of claim 1, wherein the first Vivaldi radiating element comprises a first antipodal Vivaldi radiating element and the second Vivaldi radiating element comprises a second antipodal Vivaldi radiating element.

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3. The apparatus of claim 1, wherein the first Vivaldi radiating element is configured to transmit or receive radiation with a first linear polarization; and

wherein the second Vivaldi radiating element is configured to transmit or receive radiation with a second linear polarization orthogonal to the first linear polarization.

4. The apparatus of claim 1, wherein the first Vivaldi radiating element and the second Vivaldi radiating element are configured to transmit or receive radiation with circular polarization.

5. The apparatus of claim 1, wherein the conductive layer comprises a copper conductive laminate layer.

6. The apparatus of claim 1, wherein the non-conductive support layer and the conductive layer comprise a copper clad laminate material.

7. The apparatus of claim 1, wherein the first connector is epoxied to the ground plane element and the second connector is epoxied to the ground plane element.

8. The apparatus of claim 1, further comprising a support structure secured within a third orifice in the ground plane element and supporting an edge of the first Vivaldi radiating element or the second Vivaldi radiating element.

9. The apparatus of claim 1, wherein the first Vivaldi radiating element comprises a microstrip structure on a copper clad laminate substrate.

10. The apparatus of claim 1, wherein the first solder pad comprises a microstrip structure on a copper clad laminate substrate.

11. An apparatus comprising:

a ground plane element comprising:

a non-conductive support layer,

a conductive layer arranged on the non-conductive support layer, and

a plurality of orifices through the non-conductive support layer, and the conductive layer;

a plurality of pairs of Vivaldi antenna elements arranged on the ground plane element, wherein each Vivaldi element of the plurality of pairs of Vivaldi antenna elements comprises a feed line and a solder pad, wherein each Vivaldi element of the plurality of pairs of Vivaldi antenna elements is secured to and electrically connected to the ground plane element via soldering of its respective solder pad to the

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conductive layer, and wherein the Vivaldi antenna elements of each of the plurality of pairs of Vivaldi antenna elements are arranged orthogonally and concentrically relative to each other; and

a plurality of connectors, wherein each of the plurality of connectors is arranged in a respective one of the plurality of orifices, is electrically connected to a respective one of the feed lines, and epoxy secures each of the plurality of connectors to the non-conductive support layer,

wherein the soldering and the epoxy secure the plurality of pairs of Vivaldi radiating element to the non-conductive support layer without further structural supports.

12. The apparatus of claim 11, wherein each of the pairs of Vivaldi antenna elements of the plurality of pairs of Vivaldi antenna elements comprises a pair of antipodal Vivaldi antenna elements.

13. The apparatus of claim 11, wherein the conductive layer comprises a copper laminate layer.

14. The apparatus of claim 11, wherein each of the plurality of connectors is epoxied to the non-conductive support layer.

15. The apparatus of claim 11, further comprising a cover arranged on the plurality pairs of Vivaldi antenna elements opposite to the ground plane element.

16. The apparatus of claim 15, wherein the cover is transparent to radiation sent or received by the plurality of antenna elements.

17. The apparatus of claim 15, wherein the cover comprises a closed cell foam.

18. The apparatus of claim 11, wherein each of the plurality of pairs of Vivaldi antenna elements comprises a first Vivaldi antenna element configured to transmit or receive radiation with a first linear polarization.

19. The apparatus of claim 18, wherein each of the plurality of pairs of Vivaldi antenna elements comprises a second Vivaldi antenna element configured to transmit or receive radiation with a second linear polarization orthogonal to the first linear polarization.

20. The apparatus of claim 11, wherein each of the plurality of pairs of Vivaldi antenna elements is configured to transmit or receive radiation with circular polarization.

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